

PACKAGE INSERT

Controlled Medicine
Keep out of reach of
children
Jauhi dari kanak-kanak
MAL23106038HA

1. NAME OF THE VETERINARY MEDICINAL PRODUCT

Hi-bac Solution 250 mg/mL.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 1 mL contains:

Tilmicosin (as phosphate).....250 mg

Methylparaben*.....1.2 mg

Propylparaben*.....0.24 mg

*Preservatives

3. PHARMACEUTICAL FORM

Solution for use in drinking water.

Yellow to brown clear solution.

4. CLINICAL PARTICULARS

4.1 TARGET SPECIES

Chickens (except hens producing eggs for human consumption), Turkeys, Pigs.

4.2 INDICATIONS FOR USE, SPECIFYING THE TARGET SPECIES

Pigs: For the treatment and metaphylaxis of respiratory disease in pig herds, associated with *Mycoplasma hyopneumoniae*, *Pasteurella multocida*, *Actinobacillus pleuropneumoniae* and other organisms susceptible to tilmicosin.

Chickens: For the treatment and metaphylaxis of respiratory disease in chicken flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

Turkeys: For the treatment and metaphylaxis of respiratory disease in turkey flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

The presence of the disease in the group/flock must be established before the product is used.

4.3 CONTRAINDICATIONS

Do not allow horses and other equines access to drinking water containing tilmicosin.

Do not use in case of hypersensitivity to tilmicosin or to any of the excipients.

4.4 SPECIAL WARNINGS FOR EACH TARGET SPECIES

Important: Must be diluted before administration to animals.

Pigs, chickens and turkeys: Water consumption should be monitored in order to guarantee adequate dosing. In case water consumption does not match quantities for which recommended concentrations were calculated, the concentration of the Product has to be adapted in a way that the recommended dosage will be taken up by the animals or different medication should be considered.

4.5 SPECIAL PRECAUTIONS FOR USE

• Special precautions for use in animals

For oral use only. Contains disodium edetate; do not inject.

Severely ill individuals tend to drink less and may need simultaneous treatment, preferably by parenteral medication.

Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin-related substances.

The use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria.

Medicated drinking water or milk replacer should be prepared fresh every 24 hours.

Official, national and regional antimicrobial policies should be taken into account when the product is used.

• Special precautions to be taken by the person administering the veterinary medicinal product to animals

Tilmicosin may induce irritation. Macrolides, such as tilmicosin, may also cause hypersensitivity (allergy) following injection, inhalation, ingestion or contact with

skin or eye. Hypersensitivity to tilmicosin may lead to cross reactions to other macrolides and vice versa. Allergic reactions to these substances may occasionally be serious and therefore direct contact should be avoided.

To avoid exposure during preparation of the medicated drinking water, wear overalls, safety glasses, and impervious gloves. Do not eat, drink or smoke when handling this product. Wash hands after use.

In the case of accidental ingestion, wash out mouth immediately with water and seek medical advice. In the event of accidental skin contact, wash thoroughly with soap and water. In case of accidental eye contact, flush the eyes with plenty of clean, running water.

Do not handle the product if you are allergic to ingredients in the product.

If you develop symptoms following exposure, such as skin rash, you should seek medical advice and show the physician this warning. Swelling of the face, lips and eyes or difficulty in breathing are more serious symptoms and require urgent medical attention.

4.6 ADVERSE REACTIONS (FREQUENCY AND SERIOUSNESS)

In very rare cases (less than 1 animal in 10,000 animals treated, including isolated reports), a decrease in water intake has been observed.

4.7 USE DURING PREGNANCY, LACTATION OR LAY

The safety of the veterinary medicinal product has not been established during pregnancy, lactation or lay. Use only according to the benefit/risk assessment by the responsible veterinarian surgeon.

4.8 INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

None known.

4.9 AMOUNTS TO BE ADMINISTERED AND ADMINISTRATION ROUTE

For oral use only. Do not inject.

Pigs: To be included in the drinking water to provide a daily dose of 15–20 mg/kg bodyweight for 5 days, which may be achieved by the inclusion of 200 mg tilmicosin per litre (80 mL of product per 100 litres).

Chickens and Turkeys (except hens producing eggs for human consumption): To be included in the drinking water at a daily dose of 15–20 mg/kg bodyweight in chickens and 10–27 mg/kg bodyweight in turkeys for 3 days, which may be achieved by the inclusion of 75 mg tilmicosin per litre (30 mL of product per 100 litres). One 240 mL bottle of product is sufficient to medicate 300 litres of drinking water for pigs or 800 litres of drinking water for chickens or turkeys. One 480 mL sachet of product is sufficient to medicate 600 litres of drinking water for pigs or 1600 litres of drinking water for chickens or turkeys. One 960 mL bottle is sufficient to medicate 1200 litres of drinking water for pigs or 3200 litres of drinking water for chickens or turkeys.

The uptake of medicated water/milk replacer depends on the clinical condition of the animals. In order to obtain the correct dosage, the concentration of the product should be adjusted accordingly.

To ensure a correct dosage, bodyweight should be determined as accurately as possible to avoid under-dosing.

The required dose should be measured using suitably calibrated measuring equipment.

Only sufficient medicated drinking water should be prepared to cover the daily requirements.

4.10 OVERDOSE (SYMPTOMS, EMERGENCY PROCEDURES, ANTIDOTES), IF NECESSARY

No symptoms of overdose. When pigs are offered drinking water containing 300 or 400 mg tilmicosin/litre (equivalent to 22.5–40 mg tilmicosin/kg bodyweight or 1.5–2 times the recommended concentration) commonly animals exhibit a reduced water intake. Although this has a self-limiting effect on tilmicosin intake, it could, in extreme circumstances, result in dehydration. This can be corrected by the removal of the medicated drinking water and replacement with fresh unmedicated water.

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No symptoms of overdose have been seen in chickens given drinking water containing levels of tilimicosin up to 375 mg/litre (equivalent to 75-100 mg tilimicosin/kg bodyweight or 5 times the recommended dose) for 5 days; daily treatment with 75 mg tilimicosin/litre (equivalent to the maximum recommended dose) for 10 days resulted in a reduction in faecal consistency.

No symptoms of overdose have been seen in turkeys given drinking water containing levels of tilimicosin up to 375 mg/litre (equivalent to 50-135 mg tilimicosin/kg bodyweight or 5 times the recommended dose) for 3 days; daily treatment with 75 mg tilimicosin/litre (equivalent to the maximum recommended dose) for 6 days also produced no symptoms of overdose.

4.11 WITHDRAWAL PERIOD(S)

Pigs: meat and offal: 14 days.

Chickens: meat and offal: 12 days.

Turkeys: meat and offal: 19 days.

Not authorized for use in birds producing eggs for human consumption.

Do not use within 2 weeks of the start of the laying period.

5. PHARMACOLOGICAL PROPERTIES

Pharmacotherapeutic Group: Antibacterials for systemic use, macrolides.

5.1 PHARMACODYNAMIC PROPERTIES

Tilmicosin is a semi-synthetic antibiotic of the macrolide group and is believed to affect protein synthesis. It has bacteriostatic action but at high concentrations it may be bactericidal. This antibacterial activity is predominantly against Gram-positive bacteria although some gram-negative bacteria and mycoplasmas of bovine, porcine, ovine and avian origin are also affected by this drug. In particular, its activity has been demonstrated against the following microorganisms:

➤ Pigs: *Mycoplasma hyopneumoniae*, *Pasteurella multocida* and *Actinobacillus pleuropneumoniae*.

➤ Chickens and turkeys: *Mycoplasma gallisepticum* and *Mycoplasma synoviae*. Scientific evidence suggests that macrolides act synergistically with the host immune system. Macrolides appear to enhance phagocyte killing of bacteria. Tilmicosin has been shown to inhibit in vitro the replication of the Porcine Reproductive and Respiratory Syndrome virus in alveolar macrophages in a dose dependent fashion.

Cross-resistance between tilimicosin and other macrolides and lincosycin has been observed.

Macrolides inhibit protein synthesis by reversibly binding to the 50S ribosomal subunit. Bacterial growth is inhibited by induction of the separation of peptidyl transfer RNA from the ribosome during the elongation phase. Ribosomal methylase, encoded by the erm gene, can precipitate resistance to macrolides by alteration of the ribosomal binding site. The gene that encodes for an efflux mechanism, mef, also brings about a moderate degree of resistance. Resistance is also brought about by an efflux pump that actively rids the cells of the macrolide. This efflux pump is chromosomally mediated by genes referred to as acrAB genes.

5.2 PHARMACOKINETIC PARTICULARS

Whilst blood concentrations of tilimicosin are low, there is pH-dependent macrophage accumulation of tilimicosin in inflamed tissues.

Pigs: After oral administration of 200 mg tilimicosin/l drinking water, the average active substance concentrations detected in lung tissue, alveolar macrophages and bronchial epithelium 5 days after the start of treatment were 1.44 µg/ml, 3.8 µg/ml and 7.4 µg/g respectively.

Poultry: As early as 6 hours after oral administration of 75 mg tilimicosin/L drinking water, the average active substance concentrations detected in lung and alveolar tissue were 0.63 µg/g and 0.30 µg/g respectively. 48 hours after the start of treatment, the tilimicosin concentrations in lung and alveolar tissue were 2.3 µg/g and 3.29 µg/g respectively.

Turkeys: After oral administration of 75 mg tilimicosin/L drinking water, the average active substance concentrations detected in lung tissue, air sac tissue and plasma 5

days after the start of treatment were 1.89 µg/ml, 3.71 µg/ml and 0.02 µg/g respectively. The highest mean tilimicosin concentration detected for lung tissues was 2.19 µg/g at 6 days; for air sac tissue it was 4.18 µg/g at 2 days and in the plasma it was 0.172 µg/g at 3 days.

6. PHARMACEUTICAL PARTICULARS

6.1 MAJOR INCOMPATIBILITIES

In the absence of compatibility studies, this veterinary medicinal product must not be mixed with other veterinary medicinal products.

6.2 SHELF LIFE

Shelf life of the veterinary medicinal product as packaged for sale: 2 years.

Shelf life after first opening of the immediate packaging:

·960 mL packages: 30 days.

·240 mL and 480 mL packages: 24 hours.

Shelf life after reconstitution in drinking water: 24 hours.

6.3 SPECIAL PRECAUTIONS FOR USE

Do not store above 30°C. Protect from direct sunlight.

6.4 NATURE AND COMPOSITION OF IMMEDIATE PACKAGING

·Polypropylene terephthalate (PET) bottles of 240 mL, 960 mL

·Polyethylene terephthalate pouch sachet of 480 mL

6.5 SPECIAL PRECAUTIONS FOR THE DISPOSAL OF UNUSED VETERINARY MEDICINAL PRODUCT OR WASTE MATERIALS DERIVED FROM THE USE OF SUCH PRODUCTS

Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

Veterinary medicinal product must not be disposed via waste water or the drainage systems.

Manure from treated animals should not be deposited on the same field in successive years.

7. MANUFACTURER

FARMABASE SAÚDE ANIMAL LTDA.

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Brazilian Industry

8. MARKETING AUTHORISATION HOLDER

F.E Venture Sdn. Bhd., Lot 3&5, Jalan RJS17/B,

Bandar Sunway, 46150, Petaling Jaya,

Selangor, Malaysia.

9. MARKETING AUTHORISATION NUMBER

MAL23106038HA

10. DATE OF FIRST AUTHORISATION

05-10-2023

11. DATE OF REVISION OF THE TEXT

January 2023.